

Registers copied by reference

<https://github.com/Qiskit/qiskit/issues/4177>

ROW 94

Registers copied by reference #4177

[New issue](#)[Closed](#)[#4178](#)

Cryoris opened on Apr 18, 2020 · edited by Cryoris

Edits ▾ ...

Information

- Qiskit Terra version: 0.13.0
- Python version: 3.7.5
- Operating system: MacOS Catalina

What is the current behavior?

The registers in the circuit are copied via reference. Modifying the registers in the copied circuit changes the original circuit.

Note that this causes Aqua algorithms such as the VQE to fail, if passed a quantum circuit and run on the QASM simulator. This happens because the Algorithm may add a classical register called 'c' to a copy of the Ansatz wavefunction multiple times, but since the original circuit is modified, the adding fails as a register called 'c' already exists.

Steps to reproduce the problem

```
circuit = QuantumCircuit(1)
circuit.x(0)
copied = circuit.copy()
copied.add_register(ClassicalRegister(1, name='c'))
print(circuit.cregs)
```



produces

```
[ClassicalRegister(1, 'c')]
```



What is the expected behavior?

Copy `qregs` and `cregs`.

Suggested solutions

```
# L1176 in quantumcircuit.py
cpy = copy.copy(self)
cpy.qregs = self.qregs.copy()
cpy.cregs = self.cregs.copy()
```



Cryoris added [bug](#) on Apr 18, 2020



Cryoris mentioned this on Apr 18, 2020

[↗ Properly copy registers on circuit copy #4178](#)



Cryoris added [priority: high](#) on Apr 19, 2020



Cryoris added this to the [0.14](#) milestone on Apr 19, 2020



lucian0 added this to To do in [Visualization and Jupyter Tools](#) on Apr 20, 2020



lucian0 added this to To do in [Circuits](#) on Apr 20, 2020



lucian0 removed this from To do in [Visualization and Jupyter Tools](#) on Apr 20, 2020

Assignees

No one assigned

Labels

[bug](#) [priority: high](#)

Type

No type

Projects

No projects

Milestone

0.14

Closed on Apr 30, 2020, 100% complete

Relationships

None yet

Development

[Properly copy registers on circuit copy](#)
Qiskit/qiskit

Notifications

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Participants



Properly copy registers on circuit copy #4178

<> Code

Merged mergify merged 6 commits into `Qiskit:master` from `Cryoris:4177-copy-registers` on Apr 20, 2020

Conversation 3 Commits 6 Checks 0 Files changed 2 +17 -0



Cryoris commented on Apr 18, 2020

Contributor

Summary

Fixes [#4177](#).

Details and comments

On `QuantumCircuit.copy` the registers were passed via reference and not copied. Hence changing the copied object led to changes in the original circuit.



fix copy

Unverified

9c5d8c5

Cryoris requested review from [1ucian0](#), [ajavadia](#), [ewinston](#), [kdk](#) and [mtreinish](#) as code owners 5 years ago

Cryoris added 2 commits 5 years ago

Merge branch 'master' into 4177-copy-registers

Verified

dca156a

Merge branch 'master' into 4177-copy-registers

Verified

f389173

Cryoris added `bug` `priority: high` labels on Apr 20, 2020

Cryoris added this to the **0.14** milestone on Apr 20, 2020

ajavadia previously approved these changes on Apr 20, 2020

View reviewed changes

ajavadia added the `automerge` label on Apr 20, 2020

Merge branch 'master' into 4177-copy-registers

Verified

a31f9ce

kdk requested changes on Apr 20, 2020

View reviewed changes

kdk left a comment

Member

Can you add a test covering this?



Reviewers

[kdk](#)



[ajavadia](#)



[1ucian0](#)



[ewinston](#)



[mtreinish](#)



Assignees

No one assigned

Labels

`bug`

`Changelog: Bugfix`

`priority: high`

Projects

None yet

Milestone

0.14

Development

Successfully merging this pull request may close these issues.

☒ Registers copied by reference

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3 participants

